ACTB-Associated Syndromic Thrombocytopenia (THC8): Comprehensive Disease Report

Disease: ACTB-Associated Syndromic Thrombocytopenia (ACTB-AST) **Causal gene: ACTB** (β-cytoplasmic/non-muscle actin; HGNC:132; chromosome 7p22.1) **MONDO:** MONDO:0100433 | **OMIM:** 620475 (THROMBOCYTOPENIA 8, WITH DYSMORPHIC FEATURES AND DEVELOPMENTAL DELAY; THC8) **Synonyms:** ACTB-AST; ACTB-related syndromic thrombocytopenia; Thrombocytopenia 8, with dysmorphic features and developmental delay **Category:** Mendelian, autosomal dominant

Summary

ACTB-Associated Syndromic Thrombocytopenia is caused by heterozygous missense variants in *ACTB*, the gene encoding β-cytoplasmic (non-muscle) actin. The disease-defining lesions cluster in the C-terminal 3′-coding region of the gene (exons 5 and 6). This distinguishes ACTB-AST at the molecular level from Baraitser-Winter cerebrofrontofacial syndrome (BWCFF), which is caused by missense variants elsewhere in *ACTB* (and in the paralog *ACTG1*). ACTB-AST is one of at least eight clinically distinct disorders that arise from the *ACTB*/*ACTG1* "non-muscle actinopathy" spectrum, in which the position and functional class of the variant dictate the resulting phenotype and its severity.

The core clinical triad reported in the defining cohort is **congenital thrombocytopenia with platelet anisotropy (variable platelet size), microcephaly, and mild developmental disability.** Mechanistically, the C-terminal β-actin variants are not simple loss-of-function alleles. Patient-derived fibroblasts and thrombocytes show cohort-specific alterations in β-cytoplasmic actin filament populations, including enhanced recruitment of thrombocytopenia-associated actin-binding proteins (ABPs). In megakaryocytes, this perturbed actin behavior compromises microtubule organization and thereby inhibits the final stages of platelet maturation. ACTB-AST thrombocytopenia is therefore best understood as a **production-type (biogenesis) defect**, rather than an immune-mediated platelet destruction disorder.

Because complete loss of β-actin is embryonic-lethal in mice, the human disorders are necessarily heterozygous and dominant-acting. There is no curative or disease-specific therapy; management is supportive (bleeding precautions, platelet transfusion for hemostatic challenges such as surgery, and developmental/rehabilitative therapies) together with genetic counseling. Diagnosis rests on identifying the characteristic C-terminal *ACTB* missense variant by exome/genome sequencing or inherited-thrombocytopenia/intellectual-disability gene panels, supported by a peripheral blood smear showing thrombocytopenia with platelet anisotropy. The emerging management paradigm for the entire *ACTB*/*ACTG1* spectrum is **variant-level, genotype-driven characterization**. This report is assembled primarily from aggregated, disease-level literature (OMIM, primary cohort reports, and mechanistic reviews) rather than individual EHR records; the foundational cohort is small (six patients), so frequency and natural-history estimates are provisional.

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1. Disease Information

Overview. ACTB-Associated Syndromic Thrombocytopenia (THC8) is an ultra-rare autosomal-dominant Mendelian disorder in which heterozygous missense variants in the C-terminal portion of *ACTB* (β-cytoplasmic actin) produce a syndromic form of inherited thrombocytopenia accompanied by microcephaly and mild developmental disability. The defining report described six patients with heterozygous variants clustered in the 3′-coding region of *ACTB* (exons 5 and 6), and emphasized that their clinical features are *distinct* from Baraitser-Winter cerebrofrontofacial syndrome (PMID: 30315159): *"Here, we report six patients with previously undescribed heterozygous variants clustered in the 3'-coding region of ACTB. Patients present with clinical features distinct from BWCFF, including mild developmental disability, microcephaly, and thrombocytopenia with platelet anisotropy."*

Key identifiers.

Resource	Identifier
OMIM	620475 (THC8)
MONDO	MONDO:0100433
Causal gene	*ACTB*, HGNC:132, NCBI Gene 60, UniProt P60709
Cytogenetic locus	7p22.1
ICD-10 (closest)	D69.4x (other primary thrombocytopenia) — no ACTB-AST-specific code
ICD-11 (closest)	3B64.1 (congenital/hereditary thrombocytopenia) — no specific code
Orphanet	No distinct ACTB-AST entry identified; related actinopathies under Baraitser-Winter spectrum
MeSH (closest)	Thrombocytopenia (D013921); no specific descriptor

Synonyms / alternative names. ACTB-AST; ACTB-related syndromic thrombocytopenia; Thrombocytopenia 8, with dysmorphic features and developmental delay (THC8).

Information source type. Primarily **aggregated, disease-level** evidence (OMIM, the Latham et al. primary cohort, and reviews of *ACTB*/*ACTG1* non-muscle actinopathies). Not derived from large EHR datasets; the evidence base is a small number of individually reported patients.

2. Etiology

Primary cause — genetic. The disease is monogenic: heterozygous, typically *de novo* missense variants in the 3′-coding region (exons 5–6, C-terminus) of *ACTB* (PMID: 30315159). There is no infectious or environmental cause of the primary disorder.

Genetic risk factors. The causal variant *is* the disease determinant; there are no known susceptibility loci or polygenic contributions. Within the broader *ACTB*/*ACTG1* actinopathy spectrum, the specific variant determines which of ≥8 disorders results and how severe it is (PMID: 41529692).

Environmental / lifestyle risk factors. None established. As a congenital Mendelian condition, there are no recognized toxic, occupational, dietary, or lifestyle risk factors. Advanced paternal age is a general (non-specific) consideration for *de novo* variants but has not been specifically studied here.

Protective factors. No genetic modifier alleles or environmental protective factors have been identified. The only "protective" genetic consideration is the general observation that biallelic loss of β-actin is incompatible with life (see §4, §15), constraining observed disease to heterozygous states.

Gene–environment interactions. None described. The phenotype is driven by the germline variant and its intrinsic effect on actin dynamics.

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3. Phenotypes

The defining cohort (n=6) reported a consistent syndromic combination. Frequencies below are qualitative given the small sample.

Phenotype	Type	Suggested HPO	Onset	Severity	Frequency (cohort)
Thrombocytopenia	Laboratory abnormality	HP:0001873	Congenital/ neonatal	Mild–moderate	Core / near-universal
Platelet anisotropy (variable platelet size)	Laboratory/ morphologic	HP:0011877 (anisocytosis of platelets)	Congenital	—	Core feature
Microcephaly	Clinical sign	HP:0000252	Congenital	Variable	Common
Mild developmental disability / intellectual disability	Behavioral/ developmental	HP:0001263; HP:0001249	Childhood	Mild	Common
Facial dysmorphism (reported across ACTB pLoF-type patients)	Physical	HP:0001999	Congenital	Mild	Variable

Phenotype characteristics. Onset is congenital; thrombocytopenia is present from the neonatal period and is non-progressive (a stable production defect rather than progressive marrow failure). Severity is at the milder end of the actinopathy spectrum — explicitly contrasted with the severe cortical malformations and dysmorphism of BWCFF (PMID: 30315159; PMID: 42107038). The related ACTB predicted-loss-of-function (pLoF) group shows mild developmental abnormalities and immunological symptoms, with facial dysmorphism distinct from BWCFF (PMID: 42107038).

Quality-of-life impact. No formal EQ-5D/SF-36/PROMIS data exist for ACTB-AST. Inferred impact: bleeding risk (bruising, mucocutaneous bleeding, perioperative hemorrhage) from thrombocytopenia, plus developmental/learning support needs from mild intellectual disability. Microcephaly may carry additional neurodevelopmental impact.

4. Genetic / Molecular Information

Causal gene. *ACTB* (β-cytoplasmic actin), 7p22.1; HGNC:132; NCBI Gene 60; UniProt P60709.

Pathogenic variants. - **Location/class:** Heterozygous **missense** variants clustered in the **3′-coding region (exons 5 and 6)**, encoding the C-terminus of β-cytoplasmic actin (PMID: 30315159). - **Classification:** Pathogenic / likely pathogenic by ACMG/AMP criteria (de novo occurrence, absent from population databases, clustering in a functional hotspot, concordant phenotype). - **Allele frequency:** Effectively absent from gnomAD and other population databases (private, de novo). - **Origin:** Germline; reported as **de novo** (autosomal dominant). - **Functional consequence:** Not a simple null. The C-terminal variants act through altered actin filament dynamics and aberrant actin-binding-protein (ABP) recruitment — a dominant, gain-of-abnormal-function/dominant-negative-like effect on the cytoskeleton rather than classic haploinsufficiency (PMID: 30315159).

Relationship to other *ACTB* variant classes. The C-terminal missense ACTB-AST variants are distinct from: - **BWCFF-type missense variants** (elsewhere in *ACTB*/*ACTG1*) that dysregulate actin polymerization–depolymerization and cause neuronal migration defects — the severe end of the spectrum (PMID: 41529692): *"A subset of disease-associated missense variants causes dysregulated actin polymerization-depolymerization and neuronal migration defects. In contrast, nonsense, frameshift, and missense variants enhancing protein degradation cause milder phenotypes or are benign."* - **True loss-of-function *ACTB* alleles** (deletions, nonsense, frameshift) causing a separate pleiotropic developmental disorder with reduced *ACTB* mRNA (PMID: 29220674): *"We describe heterozygous ACTB deletions and nonsense and frameshift mutations in 33 individuals with developmental delay, apparent intellectual disability, increased frequency of internal organ malformations."* - **ACTB pLoF missense/in-frame deletions** (e.g., G302A, S338-I341del) that reduce protein stability/expression and modulate cytoskeletal reorganization (PMID: 42887317).

Variant-specific phenotype illustration. The recurrent de novo *ACTB* **p.Ser348Leu** variant causes a *different* C-terminal-region syndrome — syndromic neonatal diabetes with hearing loss, neurodevelopmental features, and intestinal atresia — and notably, *"None of the 96 cases with other ACTB pathogenic variants in the Human Gene Mutation Database had diabetes"* (PMID: 42107907). This underscores that ACTB-AST is defined by *particular* C-terminal variants, not by any C-terminal variant.

Modifier genes / epigenetics / chromosomal abnormalities. None specifically established for ACTB-AST. Thrombocytopenia-associated ABPs are mechanistic partners (see §6) rather than formal modifier loci.

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5. Environmental Information

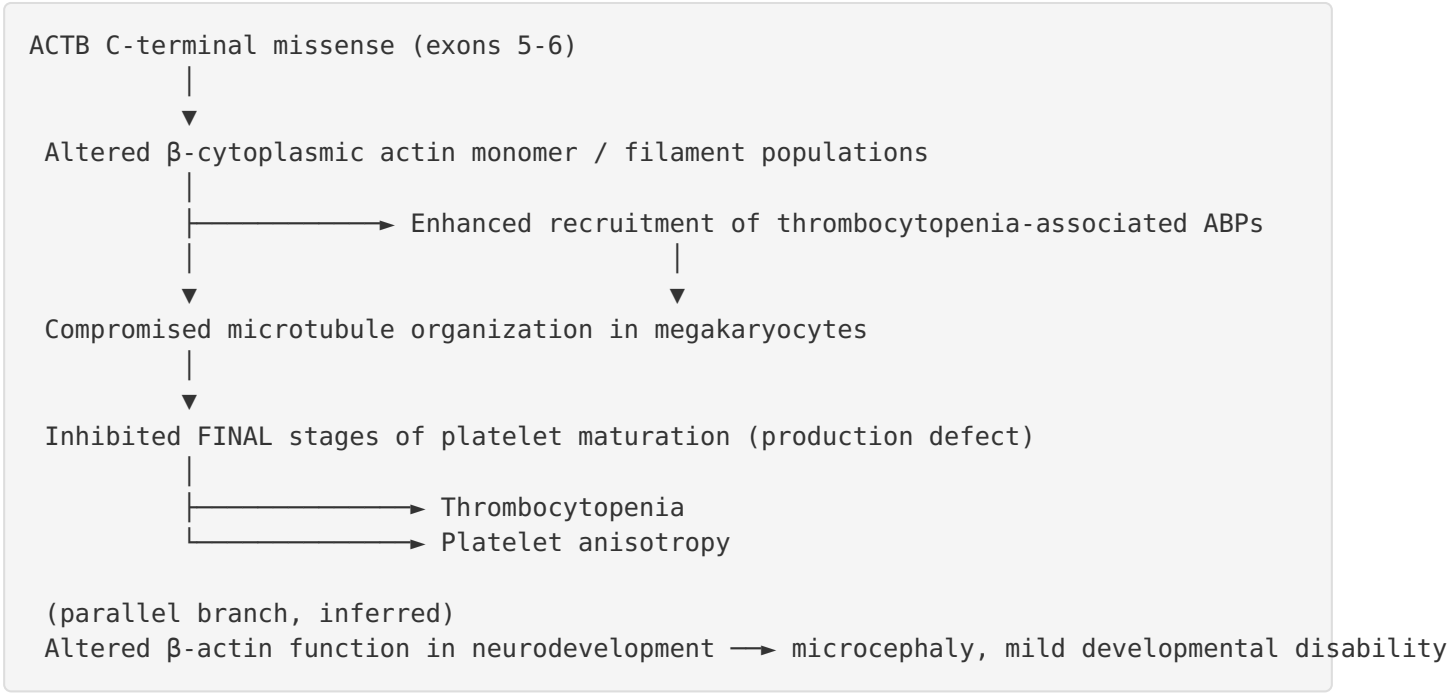
No environmental factors, lifestyle factors, or infectious agents are implicated in the causation of ACTB-AST. The disorder is a congenital, germline, monogenic condition. Environmental considerations are limited to **avoiding bleeding risk** (e.g., antiplatelet agents such as NSAIDs/aspirin, contact-injury exposure) as a management consideration rather than an etiologic factor.

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6. Mechanism / Pathophysiology

Ordered causal chain

1. A **heterozygous de novo missense variant** in the C-terminal 3′-coding region (exons 5–6) of *ACTB* **results in** production of a structurally altered β-cytoplasmic actin monomer. *(Demonstrated — variant identification.)*
2. The mutant β-actin **leads to** cohort-specific changes in β-cytoplasmic actin filament populations in patient cells (fibroblasts, thrombocytes). *(Demonstrated — patient-cell imaging + in silico modeling.)*
3. These altered filaments **result in** enhanced/aberrant recruitment of **thrombocytopenia-associated actin-binding proteins (ABPs)**. *(Demonstrated.)*
4. Perturbed actin dynamics and ABP engagement **compromise microtubule organization** within maturing megakaryocytes (actin–microtubule crosstalk is essential for platelet biogenesis). *(Demonstrated in patient megakaryocytes/thrombocytes.)*
5. Disrupted actin–microtubule coordination **inhibits the final stages of platelet maturation** (proplatelet formation/terminal platelet release). *(Demonstrated — co-examination of actin and microtubule constituents.)*
6. Impaired terminal maturation **leads to** reduced and size-variable platelet output → **congenital thrombocytopenia with platelet anisotropy** (a production-type defect). *(Demonstrated clinically.)*
7. In parallel, altered β-actin cytoskeletal function in neural/developmental tissues **contributes to microcephaly and mild developmental disability**. *(Inferred — consistent with β-actin's essential developmental roles; not fully mechanistically dissected for this variant class.)*



Mechanistic detail

- **Molecular pathways / cellular processes.** The core process is **actin cytoskeleton organization and dynamics** (GO:0030036 actin cytoskeleton organization; GO:0007010 cytoskeleton organization) coordinated with **microtubule organization** (GO:0000226) during **platelet formation** (GO:0030220) and **megakaryocyte differentiation** (GO:0030219). The general principle is established: *"Rearrangements of the microtubule (MT) and actin cytoskeleton are pivotal for platelet biogenesis. Hence, defects in actin- or MT-regulatory proteins are associated with platelet disorders in humans and mice"* (PMID: 32407474). Supporting megakaryocyte biology: Cdc42/Pak-dependent F-actin dynamics structure the demarcation membrane system (PMID: 26991240), and the spectrin-based membrane skeleton is required for proplatelet/platelet formation (PMID: 21566095).
- **Protein dysfunction.** C-terminal substitutions alter β-actin's filament behavior and its interaction surface for ABPs rather than abolishing the protein — a dominant perturbation of cytoskeletal function. Key supporting statements from the defining paper: *"these β-CYA mutations inhibit the final stages of platelet maturation by compromising microtubule organization"* and the variants cause *"the enhanced recruitment of thrombocytopenia-associated actin binding proteins (ABPs)"* (PMID: 30315159). (Contrast: pLoF variants reduce protein stability/expression — PMID: 42887317; PMID: 42107038.)
- **Cytoskeletal reorganization evidence (related variants).** ACTB pLoF variants (G302A, S338-I341del) modulate focal adhesions, cell stiffness, and cofilin/myosin-light-chain

phosphorylation in patient fibroblasts — demonstrating that C-terminal/pLoF ACTB variants reshape cytoskeletal mechanics and regulatory signaling (PMID: 42887317).

- **Immune involvement.** Thrombocytopenia here is **production-type, not immune-mediated**. (The related ACTB pLoF group can show immunological symptoms — PMID: 42107038 — but this is not the primary ACTB-AST mechanism.)
- **Subcellular localization.** Cytoplasmic actin cytoskeleton (GO:0015629 actin cytoskeleton; GO:0005856 cytoskeleton); cortical/submembranous networks of megakaryocytes and platelets.

Cell types (CL): megakaryocyte (CL:0000556), platelet/thrombocyte (CL:0000233), and fibroblast (CL:0000057, as experimental model tissue). **Anatomical process:** platelet biogenesis in bone marrow.

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7. Anatomical Structures Affected

Organ/system level. - **Hematopoietic system / bone marrow** (UBERON:0002371 bone marrow; UBERON:0000178 blood) — primary site of defective platelet production. - **Blood / circulating platelets** — site of the measurable thrombocytopenia and anisotropy. - **Central nervous system / brain** (UBERON:0000955) — microcephaly and developmental disability (secondary developmental involvement).

Tissue and cell level. - Megakaryocytes (CL:0000556) in bone marrow; platelets (CL:0000233) in peripheral blood. - Patient **fibroblasts** (CL:0000057) used as the accessible cell model demonstrating altered β-actin filament populations.

Subcellular level. - **Actin cytoskeleton** (GO:0015629) and **microtubule cytoskeleton** (GO:0005874); cortical cytoplasm of megakaryocytes/platelets.

Localization / lateralization. Systemic (not lateralized). Thrombocytopenia is a whole-blood finding; microcephaly is a bilateral/symmetric craniofacial feature.

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8. Temporal Development

- **Onset.** Congenital. Thrombocytopenia is present from the neonatal period; microcephaly is congenital; developmental disability becomes apparent through childhood. Onset pattern is **chronic/congenital**, not acute.
- **Progression.** The thrombocytopenia reflects a fixed production defect and is best characterized as **stable/non-progressive** rather than progressive marrow failure. Developmental disability is static (a developmental, not neurodegenerative, condition).
- **Disease course.** Lifelong (chronic). No relapsing-remitting pattern; no spontaneous remission expected because the defect is constitutional.
- **Critical periods.** Perioperative and traumatic/bleeding-risk windows are the principal clinically actionable periods (need for platelet support). Early childhood is the window for developmental/educational intervention.

9. Inheritance and Population

- **Inheritance pattern.** Autosomal dominant; variants reported as **de novo** (PMID: 30315159).
- **Penetrance / expressivity.** Presumed high penetrance for the carried variant; expressivity across the small cohort appears relatively consistent for the core triad, though the overall *ACTB*/*ACTG1* spectrum shows marked variant-dependent expressivity (PMID: 41529692).
- **Epidemiology. Ultra-rare.** No formal prevalence/incidence estimates exist; the defining cohort comprised six patients. No population-based registry data.
- **Why only heterozygous/dominant.** Complete β-actin knockout is embryonic-lethal in mice, so biallelic LoF is incompatible with human life, constraining all observed *ACTB* disorders to the heterozygous state (PMID: 21900491): *"We generated whole-body β-actin-knockout (Actb(-/-)) mice and demonstrated that β-actin is required for early embryonic development."*
- **Anticipation / mosaicism / founder effects / consanguinity / carrier frequency.** No genetic anticipation (not a repeat-expansion disorder). De novo origin means consanguinity and carrier frequency are not relevant; germline mosaicism is theoretically possible (documented for the allelic BWCFF — PMID: 32506774) but not specifically reported for ACTB-AST. No founder effects.

- **Demographics.** No sex predilection or ethnic/geographic clustering reported given the small numbers.

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10. Diagnostics

Recommended approach. Diagnosis is **molecular**, achieved by identifying the characteristic C-terminal (exons 5–6) *ACTB* missense variant, combined with hematologic confirmation of thrombocytopenia with platelet anisotropy.

Clinical/laboratory tests. - **Complete blood count** with platelet count: thrombocytopenia (HP:0001873; LOINC platelet count 777-3). - **Peripheral blood smear**: platelet anisotropy / variable platelet size (HP:0011877). - Mean platelet volume (MPV) and platelet morphology review support the production-type picture.

Genetic testing (GTR/GeneReviews paradigm). - **Whole-exome sequencing (WES)** or **whole-genome sequencing (WGS)** — highest-yield, able to detect the de novo missense variant and exclude phenocopies. - **Inherited-thrombocytopenia gene panels** and/or **intellectual-disability panels** that include *ACTB*. - **Single-gene *ACTB* sequencing** when the syndrome is clinically suspected. - Chromosomal microarray/karyotype/FISH are generally uninformative (no large structural lesion); mitochondrial and repeat-expansion testing are not applicable.

Clinical criteria / differential diagnosis. No formal consensus criteria. Key differentials: - **Baraitser-Winter cerebrofrontofacial syndrome** (ACTB/ACTG1 missense elsewhere; severe cortical malformation, coloboma, distinctive facies — PMID: 30315159, PMID: 27625340). - **ACTB loss-of-function pleiotropic developmental disorder** (deletions/nonsense/frameshift — PMID: 29220674). - Other **inherited thrombocytopenias** (e.g., MYH9-related disease and other giant-platelet syndromes — PMID: 18278172); other actin/ABP-related platelet disorders. - *ACTB* p.Ser348Leu syndromic neonatal diabetes (PMID: 42107907) — different C-terminal variant, different syndrome.

Screening. Cascade testing is largely moot given de novo origin; targeted variant testing can be offered to confirm de novo status and inform recurrence risk.

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11. Outcome / Prognosis

- **Survival / mortality.** No disease-specific mortality data. The thrombocytopenia is mild-to-moderate and production-type; life expectancy is not clearly reduced, though bleeding risk exists during surgery/trauma. No evidence base for 5-/10-year survival figures (not a malignancy).
- **Morbidity / function.** Chief morbidities are **bleeding tendency** (bruising, mucocutaneous and perioperative bleeding) and **neurodevelopmental** impact (mild intellectual disability, microcephaly-related).
- **Disease course / complications.** Chronic, stable course. Complications are bleeding events under hemostatic challenge and developmental/educational needs.
- **Recovery potential.** The underlying defect is not reversible; supportive measures mitigate complications. Developmental outcomes are generally in the mild range.
- **Prognostic factors.** The **specific variant** is the dominant prognostic determinant across the *ACTB*/*ACTG1* spectrum — variant class predicts severity and organ involvement (PMID: 41529692). Severity of thrombocytopenia and any additional organ features further modulate prognosis.

> **Caveat on malignancy:** Somatic *ACTB* mutations occur in certain neoplasms (e.g., tubular adenomas of the breast codon-95 hotspot — PMID: 41456879; MiT-fusion PEComas — PMID: 42331846), and AML has been reported in the allelic BWCFF (PMID: 27868373). A specific cancer predisposition for ACTB-AST is **not established**; this remains an open question.

12. Treatment

No curative or disease-specific therapy exists. Management is **supportive and genotype-informed** (PMID: 41529692; PMID: 30315159).

Supportive / hematologic. - **Bleeding precautions**; avoid antiplatelet agents (NSAIDs/aspirin) where possible. - **Platelet transfusion** for significant bleeding or hemostatic challenge (surgery, trauma, delivery) — NCIT "Platelet Transfusion" (C15360). - Antifibrinolytics (e.g., tranexamic acid) are a reasonable adjunct for mucosal bleeding by analogy to other inherited thrombocytopenias (not ACTB-AST-specific evidence).

Developmental / rehabilitative. - Early-intervention, special education, physical/occupational/speech therapy for developmental disability (NCIT "Rehabilitation Therapy" C15917; "Supportive Care" C15277).

Advanced/experimental therapeutics. None. No gene therapy, cell therapy, RNA-based therapy, or targeted therapy is available or in trials for ACTB-AST. **No ACTB-AST-specific clinical trials are registered.**

Pharmacogenomics. Not applicable (no disease-specific drug).

Treatment strategy. The authoritative framework is **variant-level, genotype-driven management** — the *ACTB*/*ACTG1* correlation work provides *"a template for the personalized genetic variant-level-driven management of the pleiotropic allelic single-gene disorders"* (PMID: 41529692).

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13. Prevention

- **Primary prevention.** None possible for a de novo germline Mendelian disorder.
- **Secondary prevention.** Early diagnosis to institute bleeding precautions and developmental support; preoperative platelet count assessment and planning.
- **Tertiary prevention.** Prevent bleeding complications (avoid antiplatelet drugs, pre-procedure platelet support); optimize developmental outcomes through early intervention.
- **Genetic counseling.** Recommended. De novo origin implies low recurrence risk for parents, but germline mosaicism cannot be fully excluded; affected individuals carry 50% transmission risk. Prenatal/preimplantation testing is feasible once the familial variant is known.
- **Immunization / public health / environmental interventions.** Not applicable.

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14. Other Species / Natural Disease

- **Taxonomy / orthologs.** *ACTB* is highly conserved. Mouse ortholog *Actb* (NCBI Gene 11461; NCBI Taxon 10090); β-actin is essential across vertebrates.
- **Natural disease in other species.** No naturally occurring animal homolog of ACTB-AST is documented in OMIA; the evidence comes from engineered mouse models (see §15).
- **Comparative biology / conservation.** β-actin's roles in cell growth, migration, and the G-actin pool are evolutionarily conserved, so disease mechanisms are expected to be conserved at the cellular level (PMID: 21900491).
- **Zoonotic potential.** Not applicable (non-infectious genetic disorder).

15. Model Organisms

- **Mouse (*Mus musculus*, NCBI Taxon 10090).** Whole-body β-actin knockout (*Actb*$^{-/-}$) mice are **embryonic-lethal**: *"We generated whole-body β-actin-knockout (Actb(-/-)) mice and demonstrated that β-actin is required for early embryonic development"* (PMID: 21900491). *Actb*$^{-/-}$ MEFs showed severe growth impairment and migration defects (not seen in γ-actin-null MEFs), with altered G-actin/F-actin ratio and changed cell-cycle/motility gene expression — establishing β-actin's unique, non-redundant cellular functions.
- **Patient-derived cellular models (in vitro).** Patient **fibroblasts** demonstrate cohort-specific β-actin filament changes and enhanced ABP recruitment (PMID: 30315159); ACTB pLoF fibroblasts (G302A, S338-I341del) reveal altered stiffness, focal adhesions, and cofilin/MLC phosphorylation (PMID: 42887317). Patient **megakaryocytes/thrombocytes** were used to demonstrate the microtubule-organization defect.
- **Model characteristics / limitations.** The embryonic-lethal full knockout cannot model the heterozygous, dominant-acting missense human disease; **knock-in of specific C-terminal variants** would be required to recapitulate ACTB-AST. No such disease-specific knock-in model is reported. Megakaryocyte culture systems (e.g., from murine fetal liver) are available tools for platelet-biogenesis studies (PMID: 22130704).
- **Resources.** MGI (*Actb*), IMPC, and standard megakaryocyte/iPSC-derived platelet culture platforms.

Mechanistic Model / Interpretation

ACTB-AST sits in the **middle of the non-muscle actinopathy severity spectrum**. The organizing principle, now formalized across *ACTB*/*ACTG1*, is that **the variant's position and functional class determine the disorder**:

Variant class	Molecular effect	Representative disorder	Severity
Missense, polymerization-dysregulating	Dysregulated actin polymerization/ depolymerization; neuronal migration defects	Baraitser-Winter CFF syndrome	Severe
Missense, C-terminal (exons 5–6)	**Altered filament dynamics; aberrant ABP recruitment; disrupted actin–MT crosstalk**	**ACTB-AST (THC8)**	**Mild–moderate, syndromic**
Specific C-terminal (e.g., p.Ser348Leu)	Variant-specific functional change	Syndromic neonatal diabetes	Variable, organ-specific
Nonsense/ frameshift/deletion (true LoF)	Reduced *ACTB* mRNA/protein	ACTB pLoF pleiotropic developmental disorder	Mild, distinct facies, immune features

The thrombocytopenia is the discriminating feature of ACTB-AST and is mechanistically a **platelet-production (biogenesis) defect**, not peripheral destruction. The mutant β-actin perturbs the actin filament populations that must coordinate with the microtubule marginal band and demarcation membrane system during terminal megakaryocyte maturation. Because *"defects in actin- or MT-regulatory proteins are associated with platelet disorders"* (PMID: 32407474), a C-terminal β-actin lesion that enhances recruitment of thrombocytopenia-associated ABPs and compromises microtubule organization logically truncates the final step of proplatelet/platelet release — yielding fewer, size-variable platelets (anisotropy).

Evidence Base

PMID	Title (abbrev.)	Role in this report
30315159	*Variants in exons 5 and 6 of ACTB cause syndromic thrombocytopenia*	**Defining paper**: variant location, phenotype, mechanism (ABP recruitment, microtubule/maturation defect)
41529692	*Molecular genotype-phenotype correlation in ACTB/ACTG1 non-muscle actinopathies*	≥8-disorder spectrum; variant-class→severity; variant-level management paradigm
29220674	*ACTB Loss-of-Function Mutations Result in a Pleiotropic Developmental Disorder*	Distinguishes true-LoF *ACTB* disorder from ACTB-AST
42887317	*Non-muscle actinopathy-associated LoF actin variants modulate cytoskeletal reorganization*	Cellular mechanism of ACTB pLoF variants (focal adhesions, cofilin/MLC)
42107907	*ACTB p.Ser348Leu in syndromic neonatal diabetes*	Variant-specificity within C-terminal region
42107038	*Overview of cytoskeletal actin mutations (review)*	Spectrum framing; pLoF group features
21900491	*β-Actin specifically controls cell growth, migration, and the G-actin pool*	Essentiality; embryonic lethality of KO; why disease is heterozygous
32407474	*Actin/microtubule crosstalk during platelet biogenesis*	General principle linking actin/MT defects to platelet disorders
26991240	*Cdc42-dependent F-actin dynamics in megakaryocytes*	F-actin role in demarcation membrane system
21566095	*Spectrin-based membrane skeleton in platelet formation*	Cytoskeletal requirement for proplatelet/platelet formation
22130704	*Visualization/manipulation of megakaryocyte cytoskeleton*	Methodological model for platelet biogenesis studies
18278172	*Congenital disorders associated with platelet dysfunctions*	Differential diagnosis context (giant-platelet/production disorders)

PMID	Title (abbrev.)	Role in this report
27625340, 27096712, 32588558, 32506774, 35054877	BWCFF clinical reports	Differential diagnosis; germline mosaicism; expressivity
27868373, 41456879, 42331846	Actinopathy/somatic-ACTB neoplasia	Open question of malignancy association

Evidence source types: human clinical cohort (30315159), human genomic/molecular review (41529692, 42107038), in-vitro patient-cell (42887317, 30315159), model organism (21900491), and megakaryocyte biology (32407474, 26991240, 21566095, 22130704).

Limitations and Knowledge Gaps

1. **Small defining cohort (n=6).** Frequency, penetrance, expressivity, and natural-history estimates are provisional; precise phenotype frequencies (e.g., % with microcephaly vs ID) are not robustly quantified.
2. **No epidemiology.** No prevalence/incidence data; no registry; no sex/ethnic/geographic data.
3. **Mechanism partly inferred.** The thrombocytopenia mechanism (ABP recruitment → microtubule disorganization → impaired terminal maturation) is directly supported, but the neurodevelopmental branch (microcephaly, ID) is inferred from β-actin's general developmental roles rather than variant-specific experiments.
4. **No ACTB-AST-specific animal model.** Only embryonic-lethal global KO exists; no knock-in of the C-terminal variants.
5. **No therapeutics or trials.** No disease-specific treatment, biomarker-driven therapy, or registered clinical trial.
6. **Malignancy risk unresolved.** Somatic *ACTB* mutations occur in some tumors and AML has been reported in allelic BWCFF, but germline ACTB-AST cancer risk is unknown.
7. **Ontology/identifier gaps.** No dedicated ICD-10/11, Orphanet, or MeSH code; mapping relies on parent "thrombocytopenia" terms.

Proposed Follow-up Experiments / Actions

1. **Expand the patient cohort** via GeneMatcher/Matchmaker Exchange to refine genotype–phenotype correlation, phenotype frequencies, and natural history.
2. **Generate variant-specific knock-in mouse and iPSC-megakaryocyte models** of the C-terminal *ACTB* variants to directly test the proplatelet-formation defect and quantify platelet output.
3. **Define the ABP interactome** of mutant vs wild-type β-actin (e.g., proximity-labeling proteomics) to identify the specific thrombocytopenia-associated ABPs recruited.
4. **Live-imaging of actin–microtubule crosstalk** in patient-derived megakaryocytes during proplatelet formation to pinpoint the maturation-arrest step.
5. **Systematically assess neurodevelopmental and microcephaly mechanisms** in neuronal models derived from patient iPSCs.
6. **Evaluate malignancy surveillance need** by aggregating longitudinal outcomes across the actinopathy spectrum.
7. **Formalize diagnostic criteria and ontology coverage** (dedicated MONDO child, HPO frequency annotations, ICD-11 extension code).

Report compiled from aggregated, disease-level literature. Primary evidence anchor: Latham et al. (2018), PMID: 30315159. Spectrum/management framework: Di Donato et al. (2026), PMID: 41529692.
